



CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours 15 minutes

12 JANUARY 2011 (a.m.)

INSTRUCTIONS TO CANDIDATES

- 1. This paper consists of THREE sections and a total of TWELVE questions. Candidates MUST answer ALL questions in all THREE sections.**
- 2. Number EACH answer correctly in the answer booklet.**
- 3. Code is to be written in the programming language, Pascal.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION I

THEORY – 60 marks

Answer ALL questions.

1. (a) A user interface allows a user to communicate with the computer to get some tasks done.
- (i) List ONE **hardware** user interface and give an example of where it is used. (2 marks)
- (ii) State TWO types of **software** user interfaces and indicate which one is more suitable for use by someone who is learning to use a computer. (3 marks)
- (b) List ONE advantage and ONE disadvantage of real-time processing. (2 marks)
- (c) Match the following applications with the type of software that BEST describes the application. In your answer booklet, write the application number (1, 2, 3) and the software type letter (A, B, C) besides it.

Application No.	Application
1	Microsoft Office 2003
2	Ubuntu Linux
3	Spanish for beginners

Software Type Letter	Type of Software
A	Operating system software
B	Specialized software
C	Integrated application software

(3 marks)

Total 10 marks

2. (a) (i) If the ASCII code for the letter 'f' is 1100110, find the ASCII code for the letter "K". Show ALL calculations. (3 marks)
- (ii) Convert the binary number 01011110 to hexadecimal. (1 mark)
- (iii) Using two's complement, find the 8-bit representation of -72. (1 mark)
- (b) John has an external hard drive with a capacity of 1 terabyte (TB) and Mary has a USB flash drive with a capacity of 16 gigabytes (GB). Calculate how many flash drives of information, similar to Mary's, can be accommodated on John's external hard drive. (2 marks)

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(c) For EACH of the following storage media, indicate whether it is a secondary or primary storage media.

(i) Flash Memory Card

(ii) Random Access Memory

(iii) Programmable Read Only Memory (PROM)

(3 marks)

Total 10 marks

3. Match the following tasks with the BEST device suitable to perform the task. In your answer booklet, write the device number (1, 2, 3 ..., 10) with the task letter (A, B, ..., J) you choose besides it.

Device No.	Device
1	Touch terminal
2	Joystick
3	Optical Mark Reader (OMR)
4	Magnetic Ink Character Reader (MICR)
5	Biometric System
6	Plotter
7	Inkjet Printer
8	Pads and Tablets
9	Headphones
10	Sensor

Task Letter	Task
A	Printing of large building plans
B	Marking a multiple choice examination
C	Preparing architectural drawings
D	Activating a buzzer when someone enters the server room
E	Making selection by touching parts of the screen
F	Listening to music on your computer
G	Allowing access to a computer by scanning the iris of your eye
H	Processing of cheques
I	Printing colour greeting cards at home
J	Used for playing games

Total 10 marks

4. A computer supplies company has eight technicians who travel to businesses and homes to sell computer supplies and accessories. Each technician has the ability to complete purchase transactions using debit or credit cards at any client's location.

- (a) Explain, giving a reason, the type of transaction media that is described above. (2 marks)
- (b) Explain what is a 'hotspot' and give ONE disadvantage of having access to a hotspot. (2 marks)
- (c) Name TWO devices that can connect to a Wi-Fi. (2 marks)
- (d) Explain the difference between 'line-of-sight communication' and 'scattered infrared communication'. (2 marks)
- (e) Explain whether line-of-sight communication or scattered infrared communication is MORE effective in transmitting data. (2 marks)

Total 10 marks

5. State ONE advantage and ONE disadvantage of EACH of the following scenarios:

- (a) Offering automated voice mail where the caller presses a number to be transferred to an extension. (2 marks)
- (b) Working at home or conducting business using a computer to communicate with colleagues and/or clients. (2 marks)
- (c) Using only credit or debit cards for transactions. (2 marks)
- (d) Use of Information Technology and computers in most areas of work. (2 marks)
- (e) Using a computer for a long time to complete an urgent task. (2 marks)

Total 10 marks

6. (a) For EACH of the file structures labelled (i) to (v) below, state whether the record arrangement is ordered (O), unordered (U) or can either be ordered or unordered (OU).

- (i) Direct Access File
- (ii) Indexed-Sequential File
- (iii) Random Access File
- (iv) Sequential File
- (v) Serial File

(6 marks)

- (b) State whether EACH of the following files is temporary or permanent:

- (i) Grandfather file
- (ii) Master file
- (iii) Transaction file
- (iv) Transaction log

(4 marks)

Total 10 marks

SECTION II

PRODUCTIVITY TOOLS – 15 marks

Answer ALL questions.

7. Answer the following questions based on the spreadsheet given below:

	A	B	C	D	E	F
1						
2	Bank	Savings	Checking	Loan	Total	
3	BD	800	500	125		
4	DB	12,000	15,000	3,000		
5	GBTI	5,000	600	200		
6	RB	15,000	1,250	900		
7	SB	2,500	1,500	600		
8						

- (a) State the range of cells containing the names of the banks. **(1 mark)**
- (b) What data format has been applied to the data with the number of accounts for each of the banks? **(1 mark)**
- (c) The data in the spreadsheet have been sorted. Indicate the name of the field used for the sort. **(1 mark)**
- (d) Write the function that should be used to find the HIGHEST number of loan accounts in a bank. **(2 marks)**
- (e) You are asked to create a **pie chart** to show the TOTAL number of accounts in EACH bank. Write the TWO ranges of data you would use. **(2 marks)**

Total 7 marks

8. A university uses a database with two tables to store information on courses that are being offered by the various faculties. The data in the two tables are given below:

TABLE 1: COURSE

Cid	Name	Credits	Faculty
MTH 200	Mathematics 2	3	NS
ECN 100	Economics 1	4	SS
HST 100	History 1	6	EH
CSI 400	Computer Science 4	8	NS

TABLE 2: FACULTY

Faculty	Dean
NS	Tiwari
SS	Scott
EH	Hussain

- (a) What field in the COURSE table is used to sort the records? **(1 mark)**
- (b) State the primary key in the COURSE table. **(1 mark)**
- (c) The two tables have been joined by a common field. State the name of the Dean of the faculty that is offering the course with Cid = ECN 100. **(1 mark)**
- (d) Write a query to show ALL courses in the NS faculty with MORE than 5 credits. **(2 marks)**
- (e) In the COURSE table it is required to insert a calculated field named FEE to compute the fee for EACH course. If the cost per credit is \$250, write the formula for the calculated field. **(2 marks)**
- (f) State the field in the COURSE table that would be MOST appropriate to use to group the data when preparing a report. **(1 mark)**

Total 8 marks

SECTION III

PROBLEM SOLVING AND PROGRAMMING – 45 marks

Answer ALL questions.

9. The two arrays below contain four numbers each.

List1:	2	4	5	8
List2:	1	3	5	7

- (a) Write code in Pascal to declare the array List1. (5 marks)
- (b) Write the contents of the two arrays after execution of the following fragment of code:

```
for x := 1 to 4 do
    List1[x] := List1[x] - List2[x]
```

 (5 marks)

Total 10 marks

10. A program reads two characters comprising *, or # into a 6-index array named TEST. Assuming that the array and all variables are already initialized write a program fragment to

- (a) initialize the array (3 marks)
- (b) print the number of stars (*) in the array (6 marks)
- (c) print the comment “NO” for all characters other than * or # in the array. (6 marks)

Total 15 marks

11. (a) Copy and complete the trace table below to show the result of the following algorithm.

```
1      last = 4
2      count = 0
3      While number = 3 to 7 Do
4          While (number <> last) DO
5              Count = Count + 1
6      End While
7      Output count
```

Number	Count
3	
4	1
5	
6	
7	

(4 marks)

- (b) What is the name of the structure from lines 3 to 6?

(1 mark)

Total 5 marks

12. Explain the difference between EACH of the following pairs of terms.

- (a) Low-level and high-level languages (2 marks)
- (b) Looping structure and conditional structure (2 marks)
- (c) Testing and debugging (2 marks)
- (d) Variable and constant (2 marks)
- (e) The symbols = and <> (2 marks)
- (f) Integer and real values (2 marks)
- (g) Pseudocode and flowchart (3 marks)

Total 15 marks

END OF TEST